

SANTIAGO MENESES GÓMEZ

Curriculum Vitae

Madrid, Spain — menegom@icloud.com — Website — LinkedIn — Github

EDUCATION & EMPLOYMENT HISTORY

Assistant Research Engineer — Image Vectorization (Adobe Collaboration Project)

Time Period: **February 2026 - Ongoing**

Institution: **Rey Juan Carlos University**

Department: **MSLab**

Location: **Madrid, Spain**

- Currently working on image processing to **vectorized** images of 3D scenes rendered with the graphic pipeline of **Mitsuba**.

Assistant Research Engineer — Virtual Clothing

Time Period: **July 2025 - February 2026**

Institution: **Rey Juan Carlos University**

Department: **MSLab**

Location: **Madrid, Spain**

- Developed an automated **end-to-end pipeline** for the generation, assembly, and processing of virtual clothing assets using **TypeScript**.
- Integrated a state-of-the-art **cloth simulation system** based on published research papers to generate high-fidelity synthetic data.
- Engineered a unified **Web Dashboard** using **React and TypeScript**, integrating the entire simulation pipeline into a single, user-friendly web application.

Master in Computer Graphics, Virtual Reality & Games

Time Period: **2024-2026**

Institution: **Rey Juan Carlos University**

Location: **Madrid, Spain**

- **Graphics API:** Developed rendering projects using **Vulkan** and **OpenGL**, gaining deep knowledge of the graphics pipeline, command buffers, and GPU memory management.
- **Advanced Rendering Techniques:** Implemented modern effects including **Screen Space Ambient Occlusion (SSAO)** and **Raytracing** (acceleration structures and intersection logic).
- **Physically Based Rendering (PBR):** Built shaders and materials following **PBR** workflows, focusing on energy conservation, Fresnel effects, and microfacet distribution functions.
- **Virtual Reality:** Developed immersive experiences, optimizing frame rates and latency for VR-specific hardware and interaction paradigms.

Gameplay Programmer (Internship)

Time Period: **Jun 2023 - Aug 2023**

Corporation: **Kumiho Esports S.L**

Department: **Programming**

Location: **Madrid, Spain**

- Led the programming effort for a **casual trivia** title, delivering a fully functional **MVP** (Minimum Viable Product) within 3 months.
- Solely responsible for the end-to-end development of the game's technical core using **C#** and **Unity**.
- Developed a robust gameplay framework, managing everything from low-level physics interactions to high-level UI systems.

Degree in Computer Informatics

Time Period: **Sept 2023 - Jan 2024**

Institution: **University of Zurich (UZH), Switzerland**

Location: **Zurich, Switzerland**

International Exchange Program (Erasmu

s+). Focused on **Advanced Software Design and Network Architectures**. Developed a foundation in

Computing Theory and Distributed Systems within a high-performance academic environment.

Degree in Videogames Design & Development

Time Period: **2020-2025**

Institution: **Rey Juan Carlos University**

Location: **Madrid, Spain**

Acquired a deep understanding of **algorithmic efficiency**, **complexity** and **software patterns**. Transitioned from general-purpose programming (**Java**, **C++**) to advanced **C#** development, specializing in **Unity**.

TECHNICAL SKILLS

- **Languages:** C++, C#, TypeScript, Java, Python, GLSL/HLSL.
- **Graphics & Engines:** Vulkan, OpenGL, Unity, Unreal Engine, PBR, Raytracing.
- **Web & Tools:** React, Node.js, Git, CMake, Docker, CUDA.

PROJECTS

OpenGL Rendering Engine

C++, OpenGL, GLSL

- Built a real-time renderer from scratch featuring a modular shader system and support for complex 3D geometries.
- Implemented advanced graphical features including **PBR (Physically Based Rendering)**, **SSAO**, and **Bloom**.

Unity Game Development Framework

C#, Unity, Git

- Developed a reusable **framework** for rapid prototyping, featuring a decoupled architecture.
- Refined several **packages**, managed through **Git**, used across multiple internal projects.

Game Jam Entries & 3D Portfolio

Unity, C#, React, TypeScript

- Participated in multiple **Game Jams (Unity)**, one showcased at **Gamescom 2023**. **Itch**
- Curated a technical **portfolio** developed in **React** in a **3D environment**.

LANGUAGES

- **Spanish:** Native.
- **English:** C1 Advanced (Cambridge Certificate).
- **German:** A1 Elementary.

ABOUT ME

Software Engineer specializing in **Computer Graphics** and **Game Development**. Experienced in building **low-level rendering engines** (OpenGL) and automated **web-based pipelines** (React/TypeScript). I am driven by the challenge of optimizing high-performance systems and creating immersive interactive experiences.